*Revision (10)-**4086***

**SIXTH SEMESTER DIPLOMA EXAMINATION**

Duration: 3 Hrs Max. Marks: 100

MODEL QUESTION PAPER

## *Communication in Instrumentation*

(Common for IT, E & I)

### <u>Part- A</u>

(Answer all 5 questions. Each question carries 2 marks)

**I** **(5 X2=10 marks)**

a) What is the advantage of electrical telemetry over other types

b) State the needs for modulation.

c) What is cladding in optical fiber?

d) Name two detectors used in fiber optic communication

e) What is Frequency Shift Keying?

### <u>Part- B</u>

(Answer any 5 questions. Each question carries 6 marks)

**II** **(5 X6=30 marks)**

a) List the advantages of current telemetry system.

b) Express Amplitude Modulated wave in frequency domain

c) Derive the expression for Frequency Modulation

d) Classify & describe optical fibers with respect to their index profile

e) List & explain different losses in optical fiber?

f) Differentiate between HART & field bus?

g) What are the HART protocol layers?

### <u>Part- C</u>

(Answer one full question from each module)

***Module I***

**III**

a) How does a general telemetry system works? Explain with diagram. 8

b) Describe the motion-balance current telemetry system 7

**Or**

**IV**

a) What are the characteristics of telemetry systems? 5

b) Describe the working of synchro transmitter and receiver 10

***Module II***

**V**

a) Explain the working of balanced modulator with circuit 7

b) Explain the FM transmitter with block diagram 8

**Or**

**VI**

a) Describe the significance of side bands in Amplitude modulation. 8

b) Explain about diode detector AM receiver 7

***Module III***

**VII**

a) Derive an expression for acceptance angle of an optical fiber. 8

b) Give the constructional details of an optical fiber 7

**Or**

**VIII**

a) With block diagram describe the general fiber optic communication system 8

b) Explain wavelength division multiplexing 7

***Module IV***

**IX**

a) Write notes on different types field-bus technologies 5

b) How do field devices are connected in a HART system 10

**Or**

**X**

a) Which are the major elements of foundation field bus 9

**b)** What is meant by blue tooth technology? 6